

Notes: Blundell, Borella, Commault, and De Nardi (AER, 2024)
Old Age Risks, Consumption, and Insurance

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1 Big picture

- **Question.** How do older households respond to income and health shocks after age 65, and what do these responses imply about insurance, medical spending, and the marginal utility of consumption?
- **Core issue.** In working-age consumption-insurance studies, a consumption response to income shocks is often interpreted as incomplete insurance. In old age, health shocks complicate this interpretation because they can affect both resources and preferences.
- **Setting.** The paper studies US households over age 65 using the Health and Retirement Study (HRS) and the Consumption and Activities Mail Survey (CAMS), which provide panel data on income, health, nondurable consumption, and out-of-pocket medical expenses.
- **Main empirical object.** The paper estimates the pass-through of transitory income and health shocks to consumption and out-of-pocket medical spending.
- **Main income result.** Transitory income shocks significantly affect nondurable consumption. The baseline pass-through is 0.127, so a 10 percent transitory income increase is associated with about a 1.3 percent increase in current nondurable consumption.
- **Main medical-spending result.** Out-of-pocket medical expenses do not significantly respond to transitory income shocks. This suggests that most older households keep out-of-pocket medical spending near a health-dependent satiation point rather than expanding it when income temporarily rises.
- **Main health result.** Transitory health shocks significantly affect both nondurable consumption and out-of-pocket medical expenses. A 1 standard deviation adverse health shock is associated with about a 2.4 percent consumption decline in the full sample and a 5.6 percent decline for lower-wealth households.
- **Main decomposition result.** Most of the consumption response to transitory health shocks comes from the marginal utility channel rather than the resources channel. In the full sample, 98.3 percent of the response is attributed to health changing the marginal utility of consumption; only 1.7 percent is attributed to the resource loss from higher medical spending.
- **Economic takeaway.** Older households face meaningful transitory risks even after Medicare and old-age social insurance. Health shocks should not be interpreted only as financial shocks; they also change how valuable consumption is in different states of health.

2 Incremental contribution of the paper

- **From working-age insurance to old-age insurance.** The paper extends the consumption-insurance literature to retirees, a group for whom health risk is central and labor-market adjustment is limited.
- **Joint treatment of income and health risks.** Rather than studying income shocks alone, the paper models income and health as parallel stochastic processes with transitory and permanent components.

- **New decomposition.** The paper develops an identification strategy that separates the consumption response to a transitory health shock into a resource effect and a marginal-utility effect.
- **Resource benchmark from income shocks.** The response of consumption to transitory income shocks identifies how a pure resource shock passes through to consumption. This multiplier is then used to evaluate how much of the health-shock response would occur through the medical-spending resource channel alone.
- **Use of CAMS spending categories.** The paper uses detailed consumption and out-of-pocket medical subcategories to show that income and health shocks affect different types of spending.
- **Evidence on health-dependent utility.** The paper provides quantitative evidence that health changes marginal utility. Luxuries respond to health shocks more than necessities, and demand-system estimates show that health reallocates spending across goods even holding total spending fixed.
- **Normative implication.** If health shocks shift marginal utility, optimal insurance is not simply about equalizing resources across states. A planner may want to give more medical goods and services, but not necessarily more nonmedical consumption, to people in worse health.

3 Data and measurement

3.1 Main data sources

- The paper combines HRS core interviews, the Consumption and Activities Mail Survey, RAND HRS files, and RAND tax calculations.
- The sample covers older US households, focuses on ages above 65, and uses biennial panel variation.
- Medicaid recipients are excluded in the main analysis because their medical-spending incentives and insurance environment differ strongly from the rest of the elderly population.

3.2 Consumption and medical expenses

- **Consumption** includes nondurable categories such as food, utilities, car-related expenses, leisure, and equipment.
- **Medical expenses** are out-of-pocket costs for drugs, medical services, and medical supplies.
- The paper also disaggregates consumption into **necessities** and **luxuries**. Necessities include food, utilities, and car-related spending. Luxuries include leisure activities and equipment.

3.3 Income

- The income measure is net income after taxes and transfers.
- It includes earnings, capital income, private pensions, Social Security and other benefits, workers' compensation, and other income.

- The use of net income means the paper measures shocks after public insurance, tax-transfer smoothing, and household self-insurance through labor supply and saving.

3.4 Health index

- Health is measured using a predicted health index based on self-reported health and objective health limitations.
- The self-reported health categories are mapped from 5 to 1: excellent, very good, good, fair, and poor.
- The paper predicts this health status using difficulties with activities of daily living and diagnosed conditions. This makes the measure less dependent on pure reporting differences.
- A higher value means better health. A one-unit change corresponds to moving by one category in self-reported health, such as from good to very good.

3.5 Wealth groups

- Net worth is assets minus liabilities, equivalized by household size.
- **Lower-wealth households** are those in the bottom quintile of the equivalized wealth distribution, below roughly US\$75,000.
- The distinction matters because lower-wealth households have less ability to self-insure and experience larger health risks.

4 Models and empirical specifications

4.1 Structural environment

The household chooses consumption c_t and out-of-pocket medical spending m_t from age 66 onward. Utility depends on consumption, total medical spending, and health:

$$\max_{\{c_t, m_t\}_{t=0}^T} E_0 \sum_{t=0}^T \beta^t s_t(\{\pi^h\}_t) u(c_t, \tilde{m}(m_t), h_t).$$

- Health affects survival probabilities and within-period utility.
- Out-of-pocket medical spending enters utility through total medical goods and services $\tilde{m}(m_t)$.
- The model allows utility to be nonseparable in consumption, health, and medical spending.

4.2 Income and health processes

The model decomposes income and health into permanent and transitory parts:

$$\ln y_t = \pi_t^y + \varepsilon_t^y, \quad \pi_t^y = \pi_{t-1}^y + \eta_t^y,$$

$$h_t = \pi_t^h + \varepsilon_t^h, \quad \pi_t^h = \pi_{t-1}^h + \eta_t^h.$$

- ε_t^y and ε_t^h are transitory income and health shocks.
- η_t^y and η_t^h are permanent income and health shocks.
- The empirical autocovariances support an MA(0) transitory component and a random-walk permanent component.

4.3 Policy functions

The household's choices can be written as:

$$c_t = c_t(a_t, \pi_t^y, \pi_t^h, \varepsilon_t^y, \varepsilon_t^h), \quad m_t = m_t(a_t, \pi_t^y, \pi_t^h, \varepsilon_t^y, \varepsilon_t^h).$$

- The pass-through of a transitory income shock captures how a pure resource shock affects consumption and medical expenses.
- The pass-through of a transitory health shock captures both resource effects and preference effects.

4.4 Pass-through coefficients

For consumption, the transitory income and health pass-through coefficients are:

$$\phi_y^c = \frac{\text{cov}(\Delta \ln c_t, \varepsilon_t^y)}{\text{var}(\varepsilon_t^y)}, \quad \phi_h^c = \frac{\text{cov}(\Delta \ln c_t, \varepsilon_t^h)}{\text{var}(\varepsilon_t^h)}.$$

The paper identifies these using future income and health growth, because future growth removes the permanent component and isolates the current transitory shock:

$$\text{cov}(\Delta \ln c_t, -\Delta \ln y_{t+1}) = \phi_y^c \text{var}(\varepsilon_t^y),$$

$$\text{cov}(\Delta \ln c_t, -\Delta h_{t+1}) = \phi_h^c \text{var}(\varepsilon_t^h).$$

The same logic gives pass-through coefficients for medical expenses, ϕ_y^m and ϕ_h^m .

4.5 Decomposition of health-shock pass-through

The paper decomposes the consumption response to a transitory health shock into a resource channel and a marginal-utility channel:

$$\phi_y^c \approx f_c^R pE[y],$$

$$\phi_h^c \approx \underbrace{\left(f_c^m \frac{dm_t}{d\varepsilon_t^h} + f_c^h \right)}_{\text{marginal utility channel}} - \underbrace{f_c^R p^m E[m]}_{\text{resources channel}} \phi_h^m.$$

- The transitory income shock identifies the resource multiplier f_c^R .
- The health shock affects resources by changing medical spending.
- The remaining response is interpreted as the effect of health on the marginal utility of consumption, including the interaction between medical spending and consumption utility.

5 Identification and empirical design

5.1 What is identified

- The paper identifies variances of transitory and permanent income and health shocks.
- It identifies pass-through coefficients from transitory shocks to consumption and medical spending.
- It decomposes the health-shock consumption response into resource and marginal-utility channels.

5.2 Moment conditions for transitory shocks

The variance of transitory income and health shocks is identified by the negative autocovariance of current and future growth:

$$\text{cov}(\Delta \ln y_t, -\Delta \ln y_{t+1}) = \text{var}(\varepsilon_t^y),$$

$$\text{cov}(\Delta h_t, -\Delta h_{t+1}) = \text{var}(\varepsilon_t^h).$$

- A current transitory shock raises current growth and reverses in the next period.
- A current permanent shock raises current growth but does not reverse in the next period.
- This is why the covariance with future growth separates transitory shocks from permanent shocks.

5.3 Key assumptions

- Income and health are well represented by random-walk permanent components plus MA(0) transitory components in biennial data.
- Transitory income and health shocks are not materially correlated in the baseline specification.
- Future shocks are unanticipated for the moment conditions used to identify pass-through.
- Transitory income shocks do not materially change the level of out-of-pocket medical expenses. The paper argues that the estimate is statistically insignificant and the dollar magnitude is small.
- Out-of-pocket medical spending affects current utility, but the model abstracts from sizable effects on future health at old ages.

5.4 Interpretation of the design

- The approach is semistructural: it is more structured than a reduced-form regression but less restrictive than fully solving and estimating the full life-cycle model.
- The identification of pass-through relies on covariance restrictions rather than on assuming that log consumption follows a random walk.
- The decomposition is strongest for transitory shocks, because transitory shocks affect current resources and current marginal utility but not future income-health distributions or future survival probabilities.

6 Tables and figures

6.1 Table 1: Covariance of current income, health, and consumption growth with current and future income and health growth

Read:

- The table reports autocovariances of income growth and health growth and cross-covariances of consumption growth with income and health growth.
- Income growth has a large positive contemporaneous variance and a negative covariance with next-period income growth. This supports a transitory income component that reverses.
- Health growth shows the same pattern: positive contemporaneous variance and negative next-period covariance.
- Consumption growth covaries with income and health changes, indicating that transitory shocks matter for household choices.

Economic message: The stochastic-process assumptions are empirically disciplined. Income and health in old age contain meaningful transitory components, not just slow permanent trends.

6.2 Table 2: Variance of the transitory and permanent income shocks

Read:

- The variance of income growth is 0.213 in the full sample.
- The variance of transitory income shocks is 0.087, implying that current transitory shocks explain about 41 percent of income-growth variance.
- The variance of permanent income shocks is 0.029, implying that current permanent shocks explain about 14 percent of income-growth variance.
- Lower-wealth households have lower estimated income-shock variance than higher-wealth households, consistent with a larger share of benefits in income.

Economic message: Retirees still face important income risk. Old-age income is not perfectly fixed, even though Social Security and pensions make income more stable than working-age earnings.

TABLE 1—COVARIANCE OF CURRENT INCOME, HEALTH, AND CONSUMPTION GROWTH WITH CURRENT AND FUTURE INCOME AND HEALTH GROWTH

	$\Delta \ln(y_t)$	$\Delta \ln(y_{t+1})$	$\Delta \ln(y_{t+2})$	$\Delta \ln(y_{t+3})$
$\text{cov}(\Delta \ln(y_t), \cdot)$	0.213 (0.007)	-0.087 (0.005)	-0.008 (0.005)	-0.002 (0.006)
$\text{cov}(\Delta \ln(c_t), \cdot)$	0.017 (0.003)	-0.011 (0.003)	-0.001 (0.004)	0.005 (0.006)
Observations	4,999	4,999	3,094	1,915
	Δh_t	Δh_{t+1}	Δh_{t+2}	Δh_{t+3}
$\text{cov}(\Delta h_t, \cdot)$	0.064 (0.002)	-0.020 (0.001)	-0.003 (0.002)	0.002 (0.002)
$\text{cov}(\Delta \ln(c_t), \cdot)$	0.005 (0.002)	-0.003 (0.002)	0.004 (0.002)	-0.004 (0.003)
Observations	4,999	4,999	3,045	1,882

Note: Standard errors in parentheses.

(a) Table 1: covariance moments for income, health, and consumption growth

TABLE 2—VARIANCE OF THE TRANSITORY AND PERMANENT INCOME SHOCKS

	All	Lower wealth	Higher wealth
$\text{var}(\Delta \ln(y_t))$	0.213 (0.007)	0.165 (0.013)	0.225 (0.008)
$\text{var}(\varepsilon_t^T)$	0.087 (0.005)	0.066 (0.009)	0.093 (0.005)
Observations	4,999	970	4,029
$\text{var}(\eta_t^P)$	0.029 (0.006)	0.017 (0.010)	0.031 (0.006)
Observations	3,401	623	2,778

Note: Standard errors in parentheses.

(b) Table 2: variance of income shocks

6.3 Table 3: Variance of transitory and permanent health shocks

Read:

- The variance of health growth is 0.064 in the full sample.
- The variance of transitory health shocks is 0.020, and the variance of permanent health shocks is also 0.020.
- Current transitory health shocks explain roughly one-third of health-growth variance.
- Lower-wealth households face substantially more health risk than higher-wealth households: transitory health-shock variance is 0.033 for lower-wealth households versus 0.017 for higher-wealth households.
- The covariance between transitory income and health shocks is small and statistically insignificant.

Economic message: Health risk is large in old age, especially for lower-wealth households. This motivates treating health as a central source of economic risk rather than a background control variable.

6.4 Table 4: Temporary health changes and the health index

Read:

- Table 4 validates the health index by relating changes in the index to changes in functional limitations.
- Difficulty walking several blocks has a large negative relationship with the health index for both household heads and spouses.
- Most limitations have negative coefficients, and none of the important significant effects have the wrong sign.
- The index therefore captures meaningful changes in physical functioning.

Economic message: The health measure is not an abstract index only. It moves in expected ways with concrete physical limitations, making the estimated health shocks economically interpretable.

	All	Lower wealth	Higher wealth
$\text{var}(\Delta h_t)$	0.064 (0.002)	0.098 (0.006)	0.056 (0.002)
$\text{var}(\varepsilon_t^h)$	0.020 (0.001)	0.033 (0.004)	0.017 (0.001)
$\text{cov}(\varepsilon_t^c, \varepsilon_t^h)$	0.002 (0.001)	0.003 (0.003)	0.002 (0.001)
Observations	4,999	970	4,029
$\text{var}(\eta_t^h)$	0.020 (0.002)	0.026 (0.005)	0.018 (0.002)
$\text{cov}(\eta_t^c, \eta_t^h)$	0.002 (0.002)	-0.003 (0.004)	0.004 (0.002)
Observations	3,401	623	2,778

Note: Standard errors in parentheses.

(a) Table 3: variance of health shocks

Change in health index	Coefficient	Standard deviation
Difficult to walk several blocks—head	-0.201	(0.016)
Difficult to walk one block—head	-0.053	(0.022)
Difficult to sit two hours—head	-0.032	(0.016)
Difficult to get up from chair—head	-0.009	(0.015)
Difficult to climb several flights stairs—head	-0.056	(0.011)
Difficult to climb one flight stairs—head	-0.103	(0.017)
Difficult to climb stoop/kneel/crouch—head	-0.049	(0.013)
Difficult to lift/carry 10 lbs—head	-0.066	(0.016)
Difficult to pick up a dime—head	0.007	(0.027)
Difficult to extend arms—head	-0.003	(0.019)
Difficult to push/pull large object—head	-0.119	(0.013)
Difficult to walk several blocks—spouse	-0.145	(0.024)
Difficult to walk one block—spouse	-0.097	(0.034)
Difficult to sit two hours—spouse	-0.054	(0.021)
Difficult to get up from chair—spouse	-0.035	(0.021)
Difficult to climb several flights stairs—spouse	-0.077	(0.016)
Difficult to climb one flight stairs—spouse	0.032	(0.024)
Difficult to climb stoop/kneel/crouch—spouse	-0.036	(0.018)
Difficult to lift/carry 10 lbs—spouse	-0.019	(0.019)
Difficult to pick up a dime—spouse	0.014	(0.043)
Difficult to extend arms—spouse	-0.085	(0.027)
Difficult to push/pull large object—spouse	-0.051	(0.016)
Observations	3,261	

(b) Table 4: health-index validation

6.5 Table 5: Pass-through estimates

Read:

- Consumption responds significantly to transitory income shocks: the full-sample pass-through is 0.127.
- The income-to-consumption response is larger for lower-wealth households at 0.202.
- Medical expenses do not significantly respond to transitory income shocks.
- Consumption responds significantly to transitory health shocks: the full-sample pass-through is 0.173.
- Medical expenses respond strongly to health shocks: the coefficient is -0.493, so worse health raises out-of-pocket medical spending.

Economic message: Income shocks look like resource shocks: they affect consumption, not much medical spending. Health shocks are different: they affect both consumption and medical expenses.

6.6 Table 6: Decomposition

Read:

- The full consumption response to a transitory health shock is 0.173.
- The resources channel contributes only 0.003, or about 1.7 percent of the total response.
- The marginal-utility channel contributes 0.170, or about 98.3 percent of the total response.
- For lower-wealth households, the resources channel is larger, at 0.018, because adverse health shocks raise their medical spending more and because a dollar resource loss has a stronger consumption effect.
- Even among lower-wealth households, the marginal-utility channel explains about 94.1 percent of the response.

Economic message: The main reason consumption falls when health worsens is not that households simply lose resources to medical bills. It is that bad health lowers the marginal utility of nonmedical consumption.

TABLE 5—PASS-THROUGH ESTIMATES

	Income shock			Health shock		
	Total	Lower wealth	Higher wealth	Total	Lower wealth	Higher wealth
Consumption ϕ_c^e	0.127 (0.036)	0.202 (0.100)	0.115 (0.038)	0.173 (0.088)	0.306 (0.132)	0.112 (0.114)
Medical expenses ϕ_m^e	0.132 (0.102)	0.234 (0.288)	0.114 (0.107)	-0.493 (0.232)	-1.171 (0.364)	-0.177 (0.286)
Observations	4,999	970	4,029	4,999	970	4,029

Note: Standard errors in parentheses.

(a) Table 5: pass-through estimates

TABLE 6—DECOMPOSITION

	All	Lower wealth	Higher wealth
Consumption ϕ_c^h	0.173 (0.088)	0.306 (0.132)	0.112 (0.114)
Resources channel	0.003 (0.002)	0.018 (0.011)	0.001 (0.002)
Change in expenses $-\phi_m^h E[mp_m]$	-1,117.91 (528.194)	-2,137.017 (690.707)	-420.397 (679.599)
Multiplier $f_{3 0}^e (10^{-6})$	3.093 (0.874)	8.382 (4.120)	2.530 (0.840)
Marginal utility channel $f_1^e \frac{dm_i}{dc_i} + f_2^e$	0.170 (0.088)	0.288 (0.132)	0.111 (0.114)
Observations	4,999	970	4,029

Note: Standard errors in parentheses.

(b) Table 6: resource versus marginal-utility decomposition

6.7 Table 7: Pass-through estimates for finer spending categories

Read:

- Transitory income shocks affect both necessities and luxuries in the full sample.
- Lower-wealth households respond to income shocks more through necessities, while higher-wealth households respond more through leisure.
- Health shocks affect luxuries much more than necessities in the full sample.
- Within luxuries, leisure activities and equipment respond strongly to health shocks.
- Among medical categories, drugs drive the response of out-of-pocket medical expenses to adverse health shocks.

Economic message: Income and health shocks move different parts of the consumption bundle. This is strong evidence that health shocks are not merely hidden income shocks: they change what households want or can enjoy.

6.8 Table 8: Budget and health elasticities for disaggregated categories

Read:

- Food, utilities, and car spending have budget elasticities below one, supporting their interpretation as necessities.
- Leisure and equipment have budget elasticities above one, supporting their interpretation as luxuries.
- Health improvements increase the budget share of leisure and car-related spending but reduce the shares of food, utilities, and equipment.
- The demand-system estimates therefore show that health reallocates spending across goods even holding total spending fixed.

Economic message: Health changes marginal utility across categories. Sick and healthy households do not simply spend different amounts; they allocate spending differently.

TABLE 7—PASS-THROUGH ESTIMATES

	Income shock			Health shock		
	Total	Lower wealth	Higher wealth	Total	Lower wealth	Higher wealth
Consumption ϕ_c^*	0.127 (0.036)	0.202 (0.100)	0.115 (0.038)	0.173 (0.088)	0.306 (0.132)	0.112 (0.114)
Necessities	0.109 (0.038)	0.314 (0.111)	0.075 (0.040)	0.076 (0.090)	0.344 (0.141)	-0.046 (0.112)
Food	0.090 (0.062)	0.425 (0.183)	0.033 (0.065)	0.045 (0.152)	0.697 (0.266)	-0.259 (0.183)
Utilities	0.099 (0.053)	0.223 (0.128)	0.077 (0.057)	0.044 (0.128)	-0.127 (0.191)	0.125 (0.165)
Car-related	0.098 (0.046)	0.248 (0.125)	0.073 (0.050)	0.285 (0.117)	0.580 (0.184)	0.147 (0.148)
Luxuries	0.110 (0.063)	-0.186 (0.178)	0.160 (0.066)	0.366 (0.150)	0.206 (0.220)	0.438 (0.193)
Leisure activities	0.219 (0.091)	-0.190 (0.333)	0.290 (0.086)	0.426 (0.231)	0.180 (0.372)	0.536 (0.282)
Equipment	0.023 (0.068)	-0.297 (0.175)	0.077 (0.073)	0.401 (0.159)	0.115 (0.228)	0.536 (0.203)
Medical expenses ϕ_m^*	0.132 (0.102)	0.234 (0.288)	0.114 (0.107)	-0.493 (0.232)	-1.171 (0.364)	-0.177 (0.286)
Drugs	0.063 (0.109)	0.134 (0.285)	0.050 (0.117)	-0.619 (0.248)	-0.936 (0.409)	-0.472 (0.304)
Services and supplies	-0.021 (0.144)	-0.024 (0.403)	-0.022 (0.152)	0.098 (0.343)	-0.173 (0.524)	0.222 (0.433)
Observations	4,994	966	4,028	4,994	966	4,028

Note: Standard errors in parentheses.

TABLE 8—BUDGET AND HEALTH ELASTICITIES, FOR DISAGGREGATED CATEGORIES

	Food	Utilities	Car	Leisure	Equipment
Budget shares	0.271 (0.007)	0.232 (0.006)	0.159 (0.005)	0.208 (0.007)	0.131 (0.005)
Budget elasticities	0.777 (0.023)	0.577 (0.027)	0.797 (0.030)	1.864 (0.033)	1.086 (0.035)
Health elasticities					
Whole sample	-0.117 (0.025)	-0.091 (0.029)	0.104 (0.032)	0.324 (0.035)	-0.235 (0.038)
Lower wealth	-0.121 (0.023)	-0.096 (0.027)	0.200 (0.033)	0.488 (0.067)	-0.203 (0.041)
Higher wealth	-0.107 (0.027)	-0.207 (0.031)	0.058 (0.033)	0.345 (0.034)	-0.084 (0.039)

Note: Standard errors in parentheses.

(a) Table 7: disaggregated pass-through estimates

(b) Table 8: budget and health elasticities

7 Short summary

- The paper studies income risk, health risk, consumption, and out-of-pocket medical spending among US households over age 65.
- It uses HRS and CAMS data to estimate transitory and permanent components of income and health.
- Both income and health have substantial transitory components in old age.
- Transitory income shocks significantly affect nondurable consumption but do not significantly affect out-of-pocket medical expenses.
- Transitory health shocks significantly affect both consumption and medical expenses.
- A 10 percent transitory income shock changes consumption by about 1.3 percent in the full sample.
- A 1 standard deviation adverse health shock lowers consumption by about 2.4 percent in the full sample and by about 5.6 percent among lower-wealth households.
- Adverse health shocks raise medical expenses substantially, especially among lower-wealth households.
- The consumption response to health shocks is almost entirely due to the marginal-utility channel, not the resource channel.
- Disaggregated spending patterns and demand-system estimates support the view that health affects preferences over consumption categories.

8 Conclusion

8.1 Core conclusion

Blundell, Borella, Commault, and De Nardi show that older households face substantial transitory risks in both income and health. Consumption is not fully insured against transitory income shocks, and health shocks affect not only financial resources through medical expenses but also the ability to enjoy consumption. The central result is that adverse health shocks reduce consumption mainly because they lower the marginal utility of nonmedical consumption.

8.2 Economic intuition

The income-shock result follows the familiar incomplete-insurance logic: when income temporarily rises or falls, households do not fully smooth consumption. The health-shock result is different. A health shock can raise medical needs and thereby reduce resources, but the estimated resource channel is small. Instead, bad health appears to make many nonmedical consumption goods less valuable or less usable. This is why luxuries, especially leisure-related spending, respond to health shocks more than necessities.

8.3 Incremental contribution revisited

The paper advances the literature in three connected ways. First, it brings consumption-insurance methods to old age, where health shocks are first-order. Second, it provides a new decomposition that separates resource effects from marginal-utility effects. Third, it uses detailed spending categories to show that the mechanism is visible in the composition of consumption, not just in total spending.

8.4 Implications for insurance and policy

The results suggest that old-age insurance analysis should include both income and health risks. Medicare and other programs insure parts of medical spending, but the paper shows that households still face sizable transitory shocks and that health changes the value of resources across states. A planner insuring a pure income shock wants to smooth resources. A planner insuring a health-related marginal-utility shock may want to reallocate resources differently: more medical goods and services in bad health, but not necessarily more nonmedical consumption.

8.5 Limitations

The analysis depends on the assumed income and health shock processes and on biennial HRS/CAMS timing. The health index is constructed from survey measures, so it is informative but not a clinical health measure. The medical-spending variable captures out-of-pocket spending, not total medical care. The decomposition is semistructural and focuses on transitory shocks; permanent health shocks would operate through additional channels such as survival probabilities and future income-health distributions. Finally, the paper discusses normative implications qualitatively rather than solving a full quantitative optimal-insurance model.

8.6 Takeaway

The enduring takeaway is that health shocks in old age are not just income shocks with medical bills attached. They reshape the marginal utility of consumption. This changes how economists should interpret consumption declines after health shocks and how insurance systems should think about protecting older households.